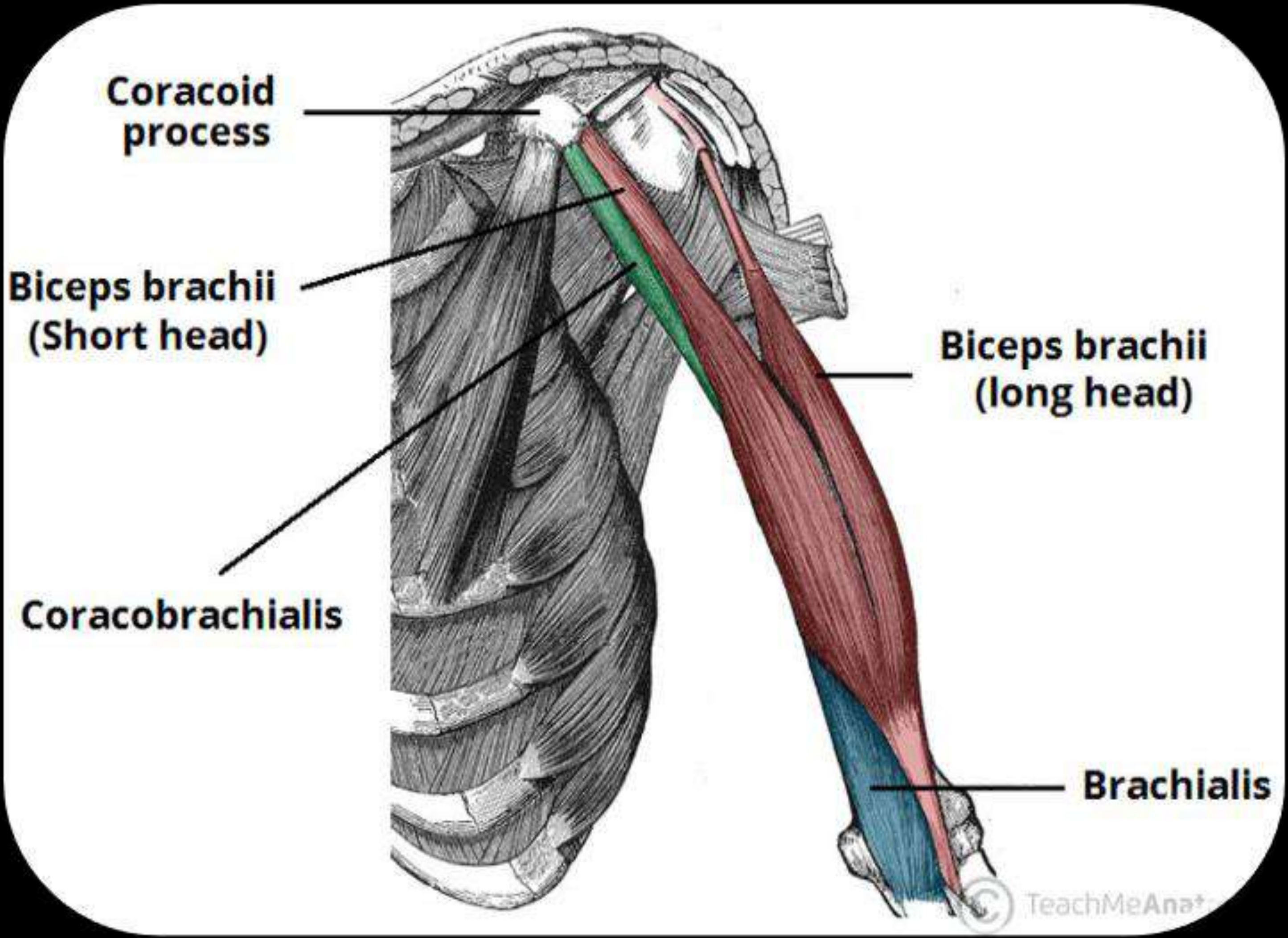
The Upper Limb

The Arm



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CONTENTS OF THE LECTURE:

Muscles of the Arm

- ✓ Anterior compartment
- ✓ Posterior compartment

Cubital fossa..

Place, Boundaries and contents.

The Brachial artery:

- ✓ Its course, branches, main relations, Profunda brachii
- ✓ Site of pulsation.

The Nerves of the arm:

Musculocutaneous, median, radial and ulnar nerves.

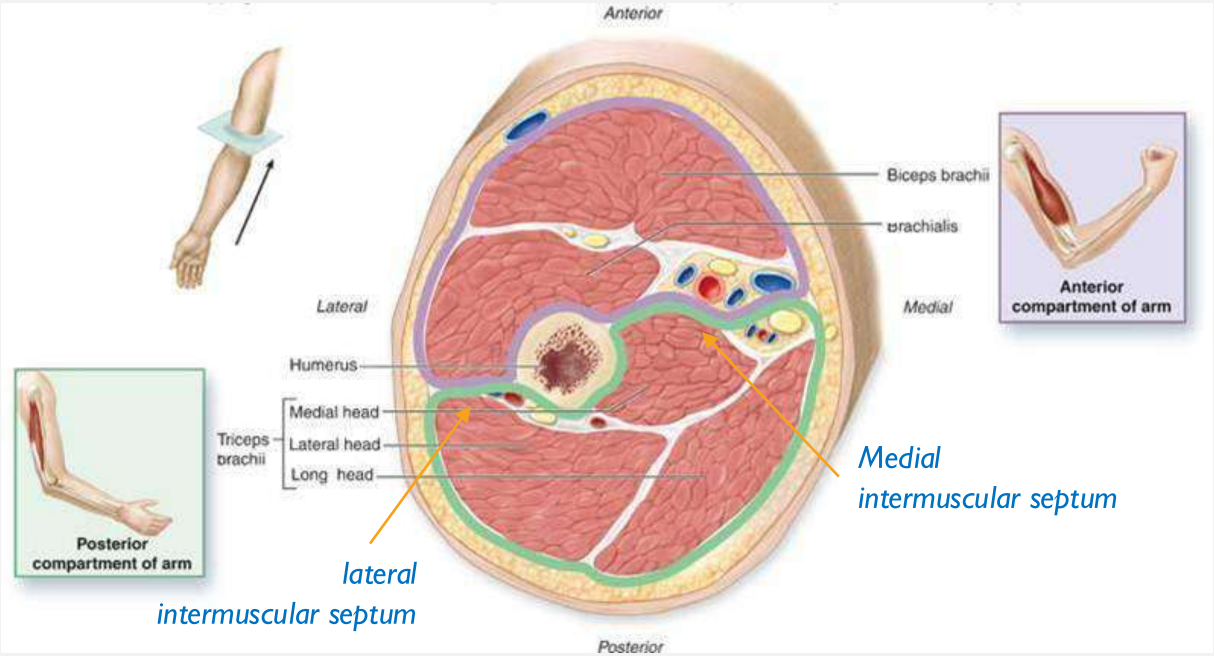
The superficial veins of ‘upper limb and its clinical importance.

The fascial compartments of the arm

The upper arm is surrounded by a sheath of deep fascia.
Medial & lateral intermuscular septa,
extending from the sheath and attached to the
medial & lateral supracondylar ridges of 'humerus.

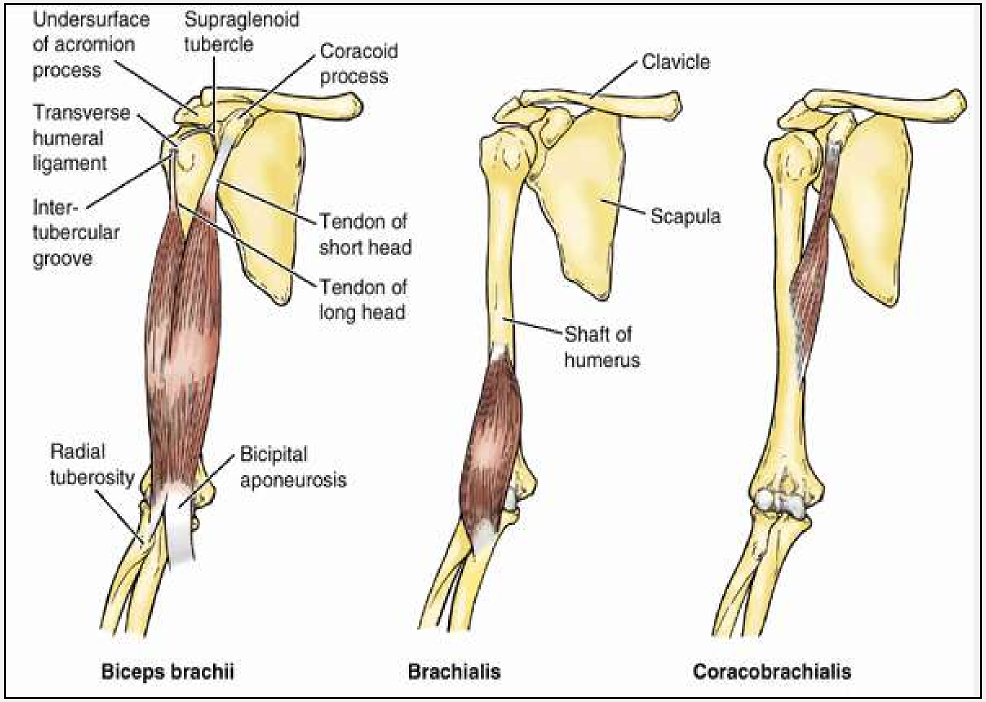
So, the upper arm is divided into:
Anterior & Posterior fascial compartments, each having its muscles, nerves, &arteries.

https://www.youtube.com/watch?v=rUq6-WXMik0



Muscles of the Arm

Anterior compartment



- Muscles: Biceps brachii, Coracobrachialis, Brachialis
- Blood supply: Brachial artery
- Nerve supply: Musculocutaneous n.

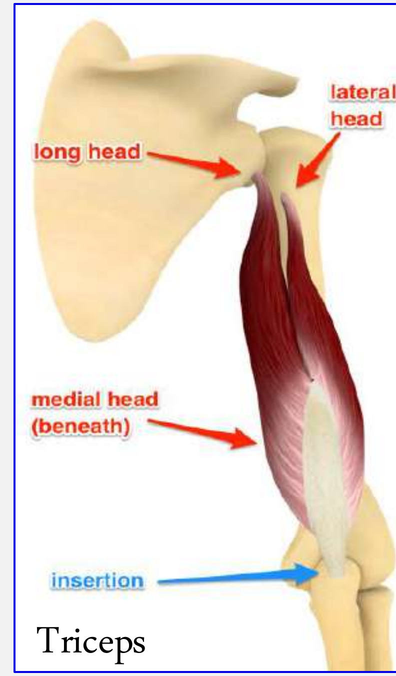
➢ **Other Structures:**

Median n.

Ulnar n.

Radial n. (present in the lower part of 'compartment)

Posterior compart



- Muscles: Triceps
- Blood supply:
  Profuna brachii artery
  Ulnar collateral artery
- Nerve supply: Radial nerve

➢ **Other Structures:**

Radial n. (in the upper part)

Ulnar n. (in the lower part)

BICEPS BRACHII

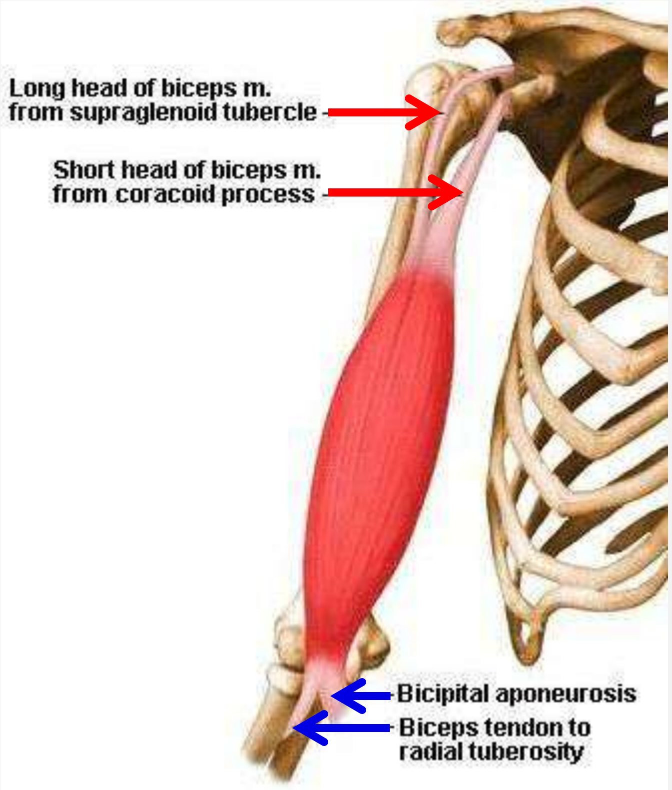
Origin:

Long head:

supraglenoid tubercle of the scapula.

Short head:

coracoid process of the scapula.



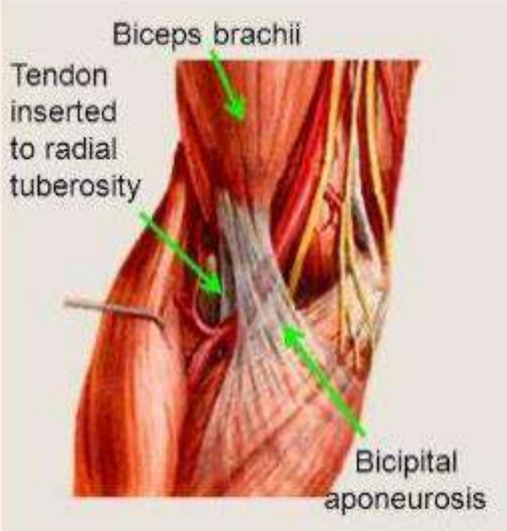
Insertion:

By tendon → inserted into ‘radial tuberosity.

By bicipital aponeurosis→ inserted into the deep fascia of the forearm.

Nerve Supply:

Musculocutaneous nerve



Action:

- Weak flexor of Shoulder J. (arm)
- Flexion of the Elbow Joint.(forearm)
- Supination of the forearm

CORACOBRACHIALIS

Origin: Coracoid process of the scapula

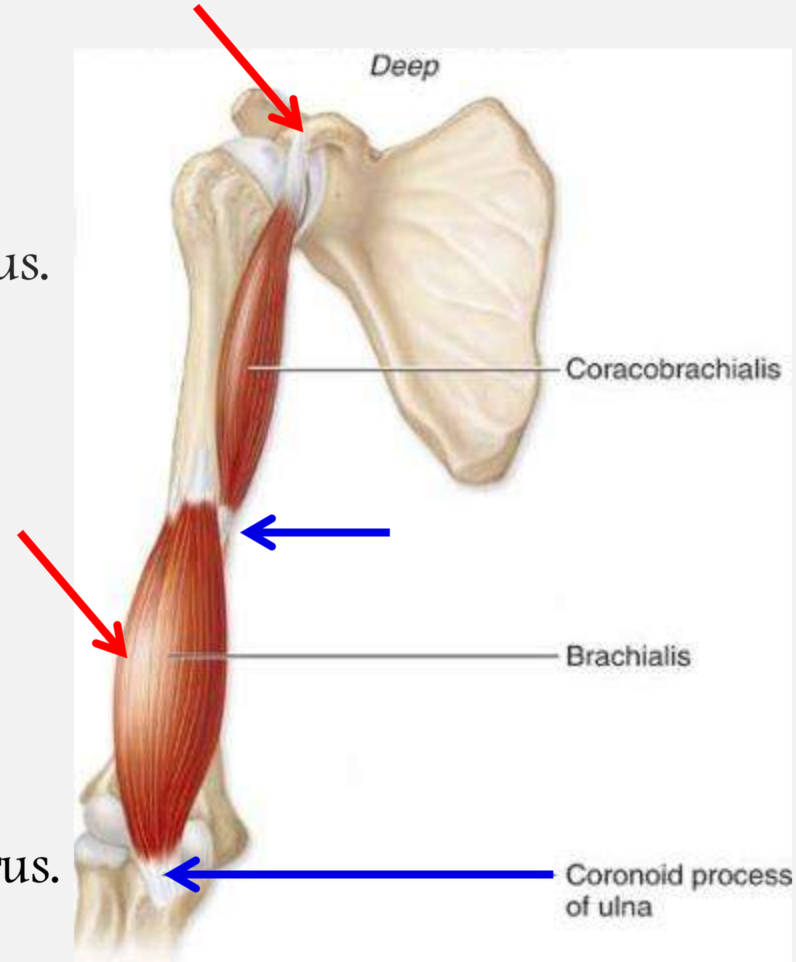
Insertion:

the middle third of the medial side of the humerus.

Nerve Supply: Musculocutaneous nerve

Action:

- Flexion of the Shoulder J.
- Weak adduction of the Shoulder J.



BRACHIALIS

Origin: Anterior parts of lower half of the humerus.

Insertion:

The coronoid process of ulna (ulnar tuberosity).

Nerve Supply: Musculocutaneous nerve

Small lateral part by Radial n.

Action:

- Flexion of the Elbow J.

TRICEPS

Origin:

A. Long head: infraglenoid tubercle of the scapula.

B. Lateral head: upper half of the back of the humerus, **above and lateral** to the spiral groove.

C. Medial head: lower half of the back of the humerus, **below and medial** to the spiral groove.

Spiral/Radial groove

Insertion:

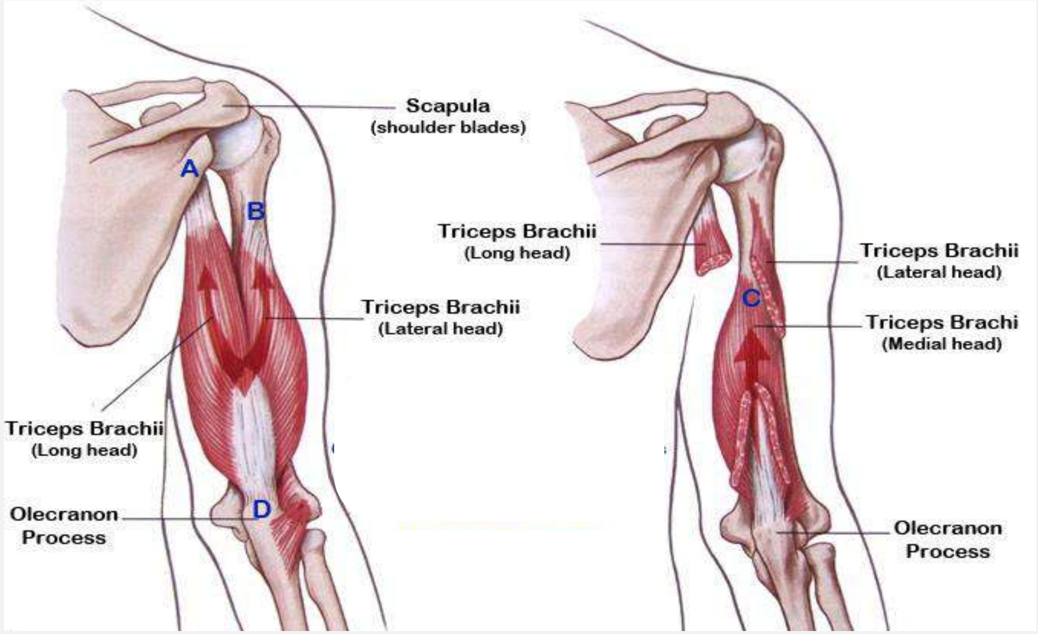
D: The olecranon process of ulna.

Nerve Supply:

Radial nerve

Action:

Extension of the elbow joint.

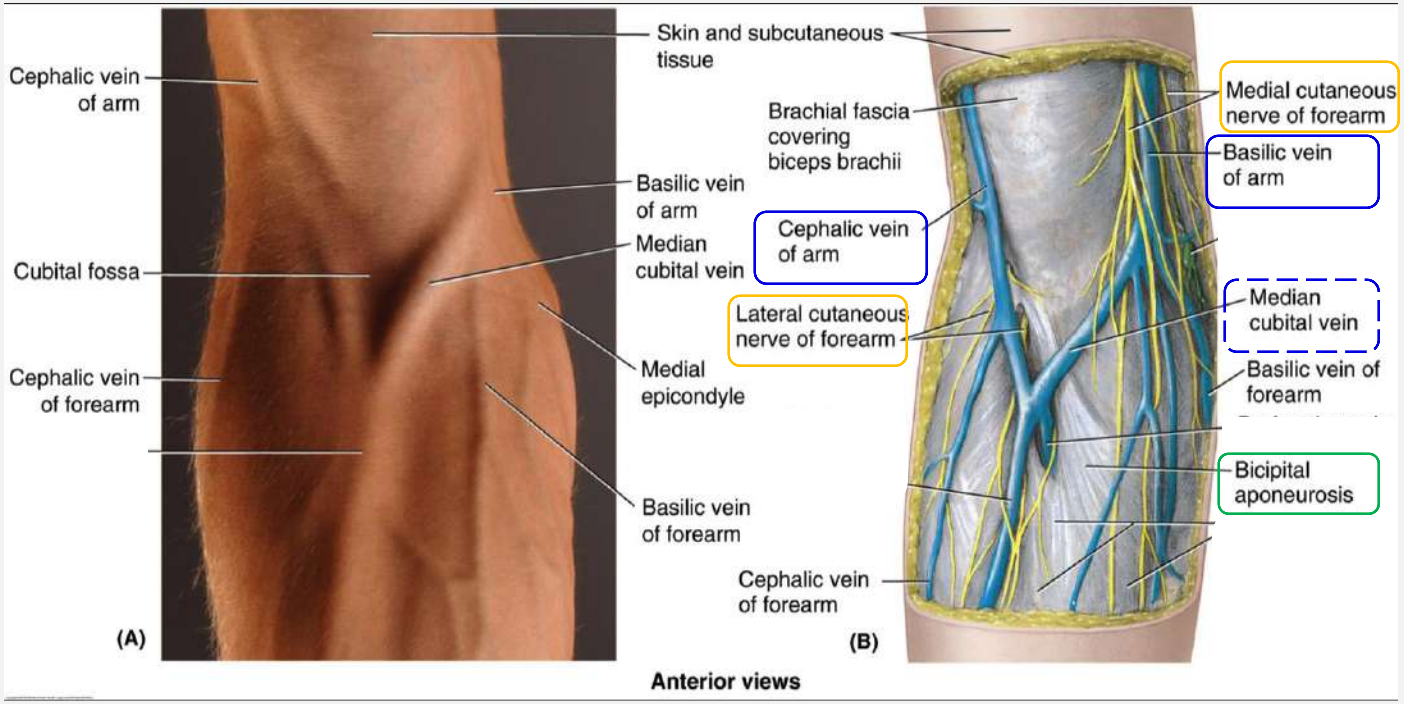


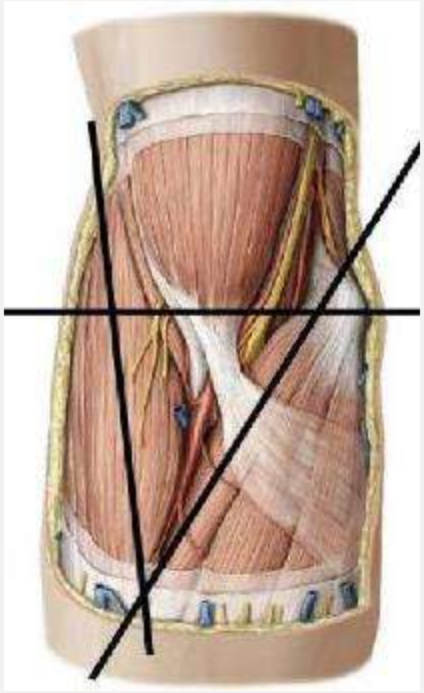
The Cubital Fossa

Site:

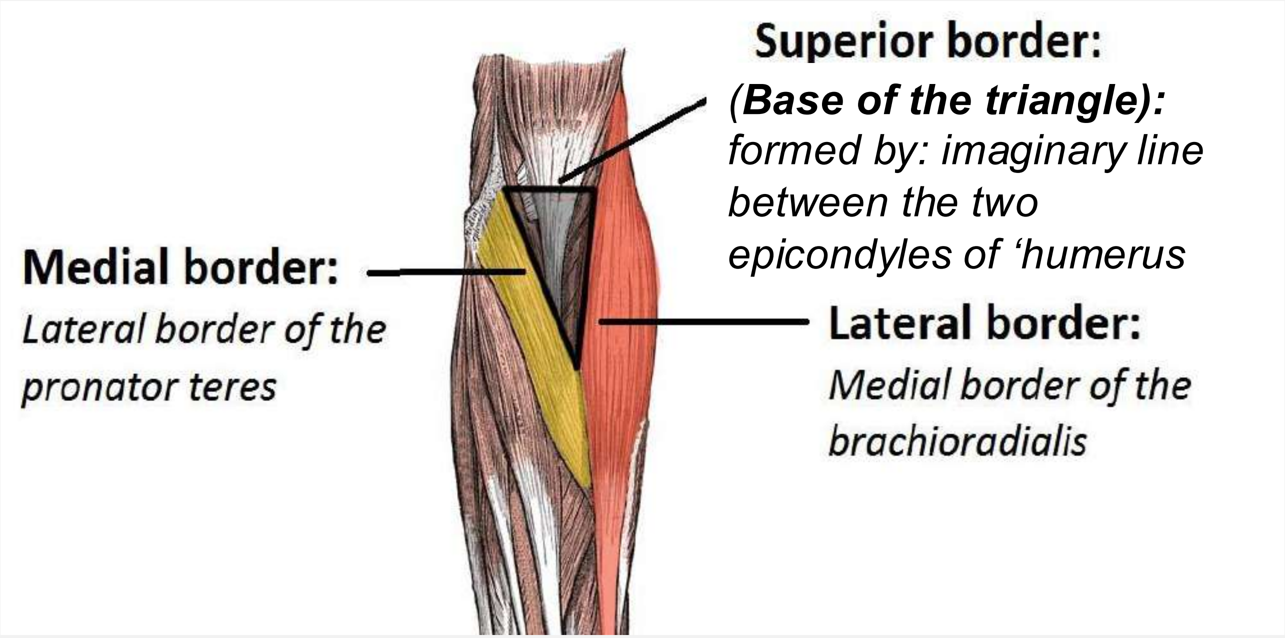
It is a triangular depression in front of the elbow joint.

- Boundaries
- Roof
- Floor
- Contents





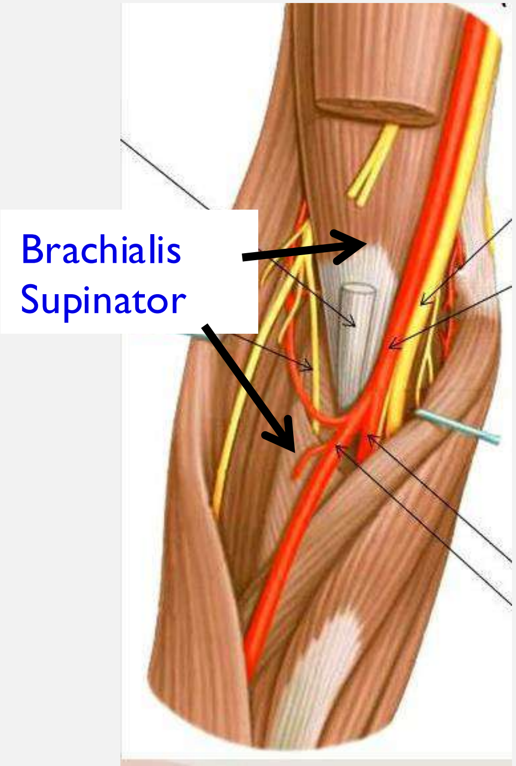
Boundaries



The **Roof** is formed by:
skin & fascia
(containing median cubital vein,
medial & lateral cutaneous ns. of forearm)
It is reinforced by 'bicipital aponeurosis.

The **Floor** is formed by two muscles:

Supinator laterally
Brachialis medially.



Contents: (from medial → lateral side)

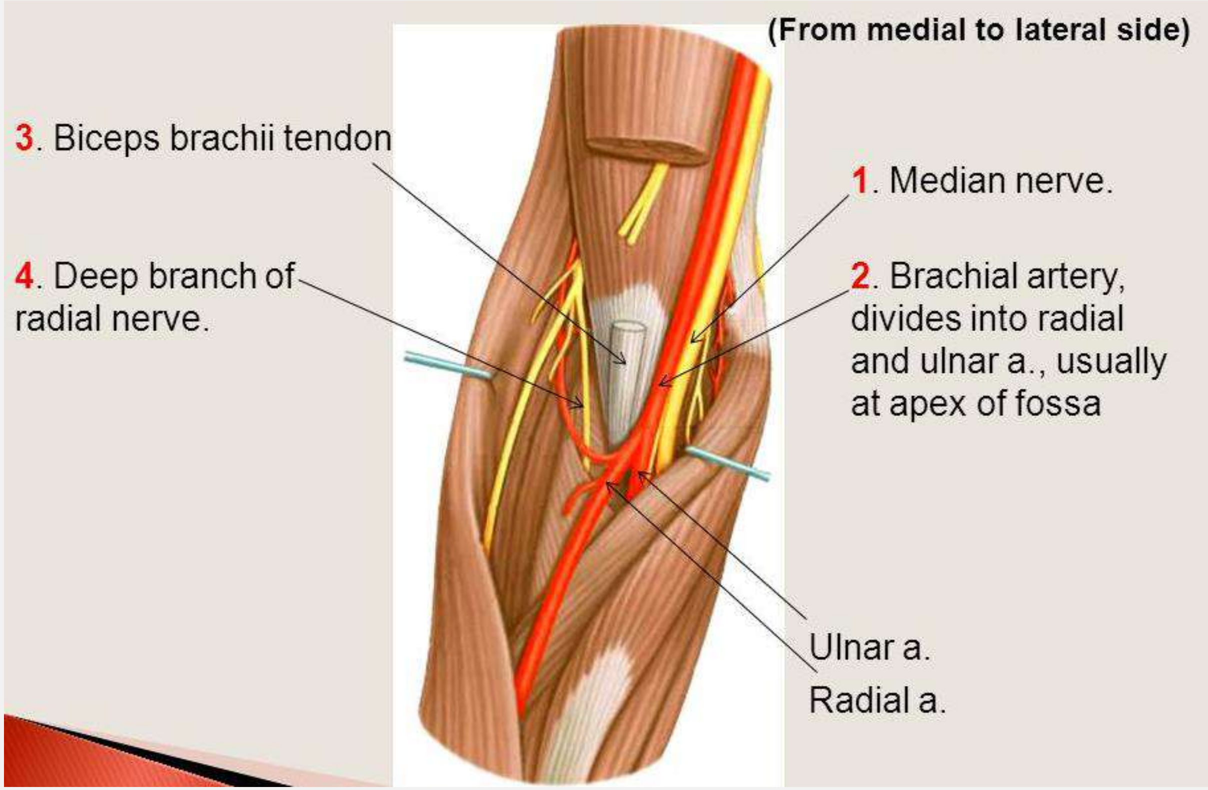
The median nerve

The 'brachial artery & its bifurcation into ulnar & radial arteries,

The Biceps brachii tendon

The radial nerve & its deep branch.

The supratrochlear lymph node: in the upper part of 'fossa.



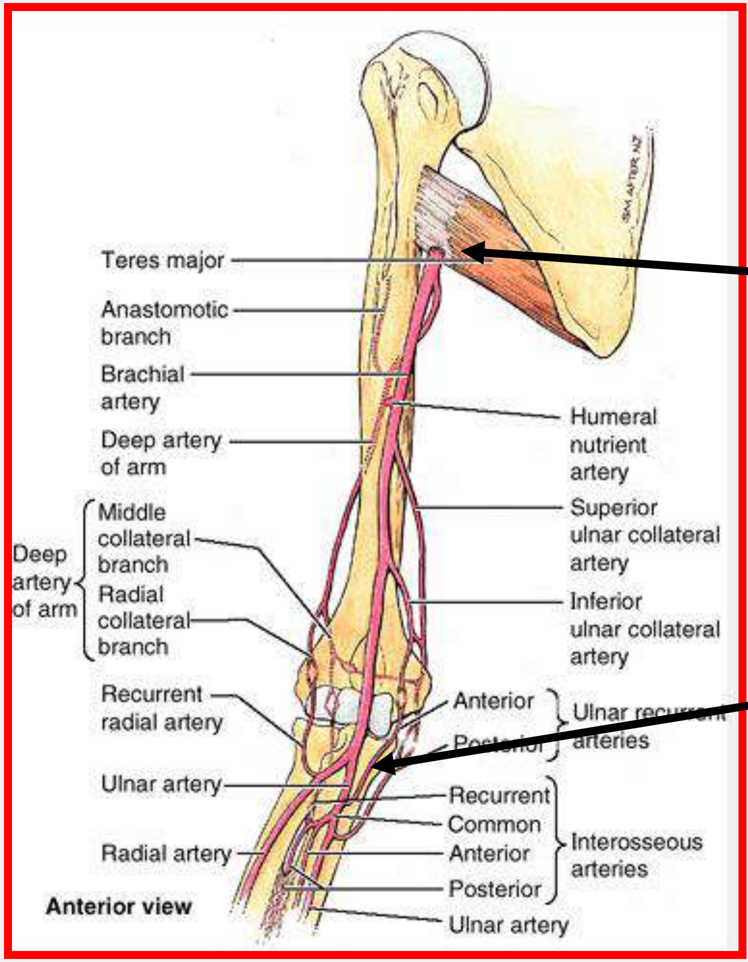
WHAT IS THE CLINICAL IMPORTANT OF THE CUBITAL FOSSA

- Phlebotomy – venous blood drainage from median cubital vein
- Blood pressure measurement - stethoscope is placed over the brachial artery in the cubital fossa. The brachial pulse may be palpated in the cubital fossa.

The Brachial artery

Begins
Ends
Branches
Relations

The artery of the upper arm:

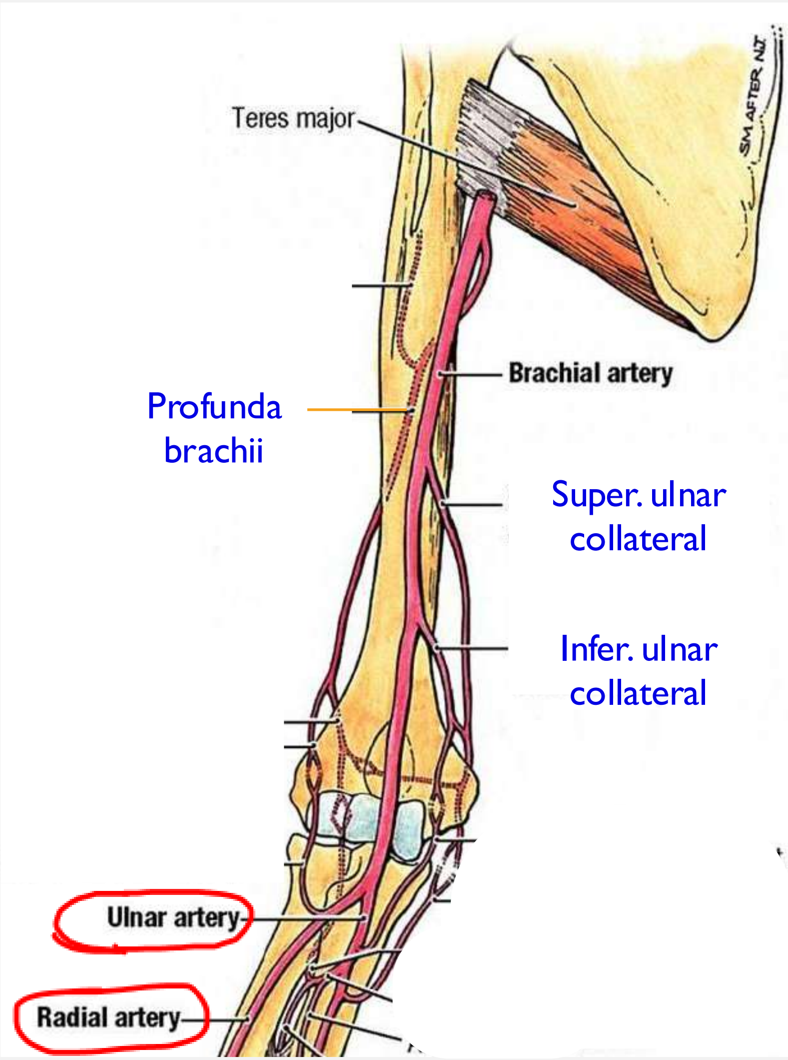


Begins

at the lower border of teres **major** ms.
as a continuation of the axillary artery.

Ends

opposite to the neck of the radius
by dividing into radial & ulnar arteries.



Branches:

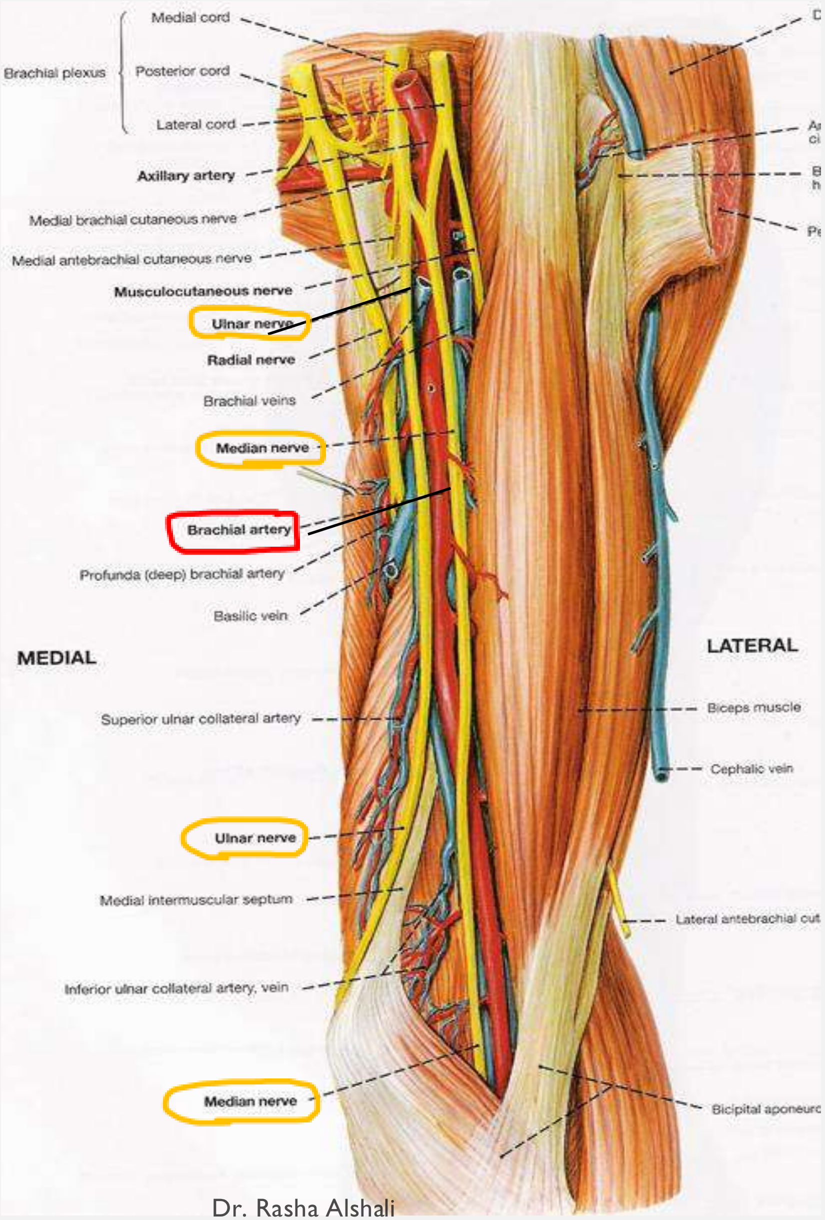
- **Profunda (deep) brachii artery:**
 arises near 'beginning of 'brachial artery.
accompanies the radial nerve in the spiral groove of ' humerus.
- **Superior ulnar collateral artery:**
 arises in 'middle of 'artery
& follows 'ulnar nerve
- **Inferior ulnar collateral artery :**
arises near 'end of 'artery

They shares in '**anastomosis** around 'elbow J.

- **Muscular branches:**
Supply muscles of ant. compartment of 'arm.
- **Nutrient artery:**
Supplies the humerus

- **Terminal branches:**

Radial artery & Ulnar artery



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Relations:

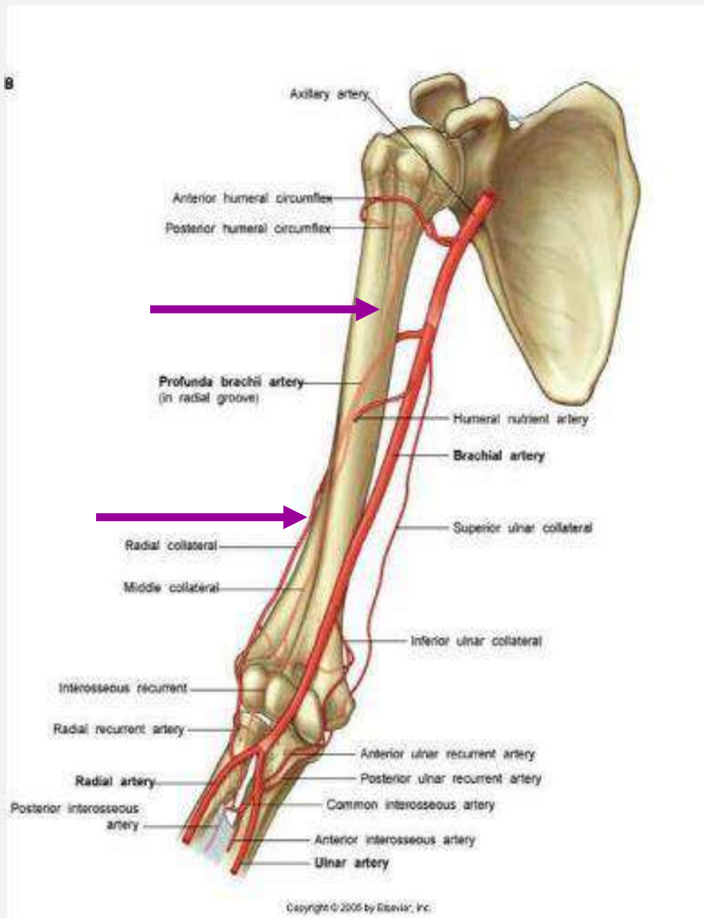
- **Anteriorly:** it is superficial & overlapped by coracobrachialis & biceps.

- **Posteriorly:**
the coracobrachialis, brachialis & triceps.

- The ulnar nerve → medially
- The median nerve was lateral then it crosses in front of the middle part of 'brachial artery to be medial.

PROFUNDA BRACHII ARTERY
= DEEP BRACHIAL @

Begins: arises from the brachial artery near its beginning.
Course: it turns backward & accompanies the radial nerve in the spiral groove of the humerus.



Branches:

- ➢Muscular branches to the triceps
- ➢Ascending branch
  anastomoses with circumflex humeral arteries around surgical neck of 'humerus.
- ➢Descending branches
  Share in anastomosis around the elbow J.

-Nerves in the arm:

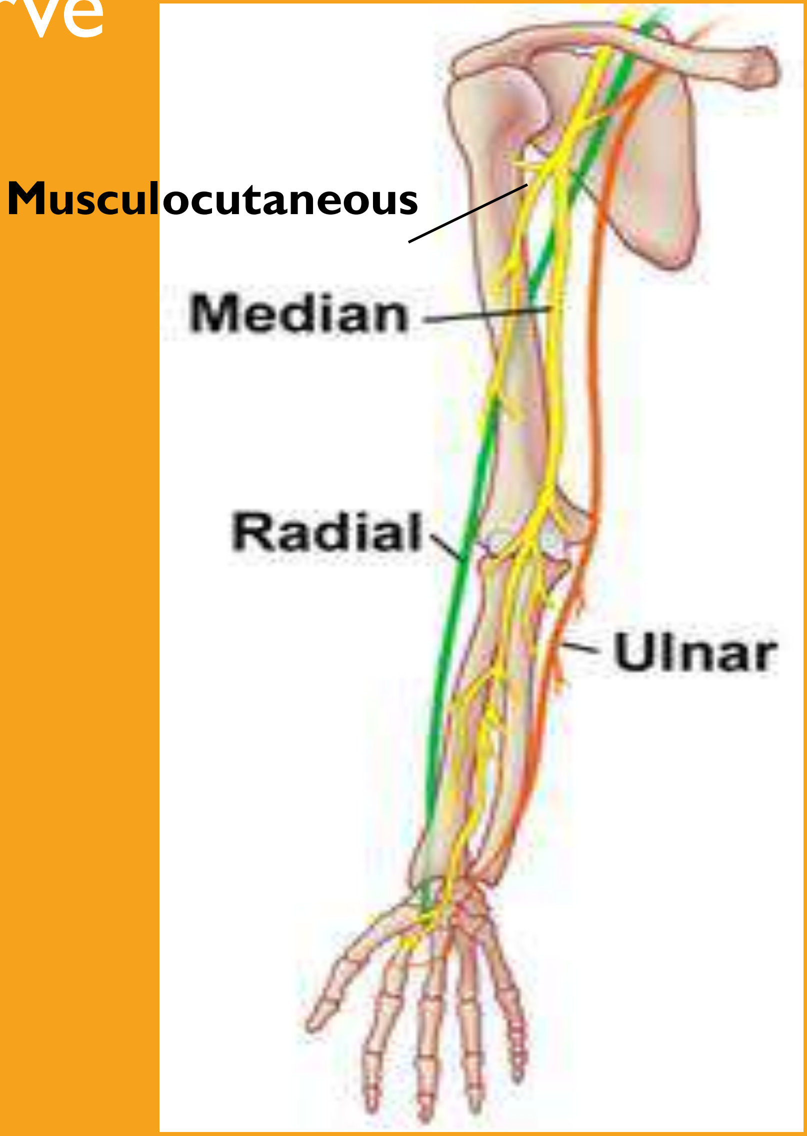
- Musculocutaneous nerve
- Median nerve
- Ulnar nerve
- Radial nerve

Origin

Course in arm

Branches:

1. Muscular
2. Cutaneous
3. Articular



Musculocutaneous Nerve

Begins:
in the axilla, arises from the lateral cord of ‘brachial plexus.
It runs downward and pierces ‘coracobrachialis
& descends between biceps & brachialis.

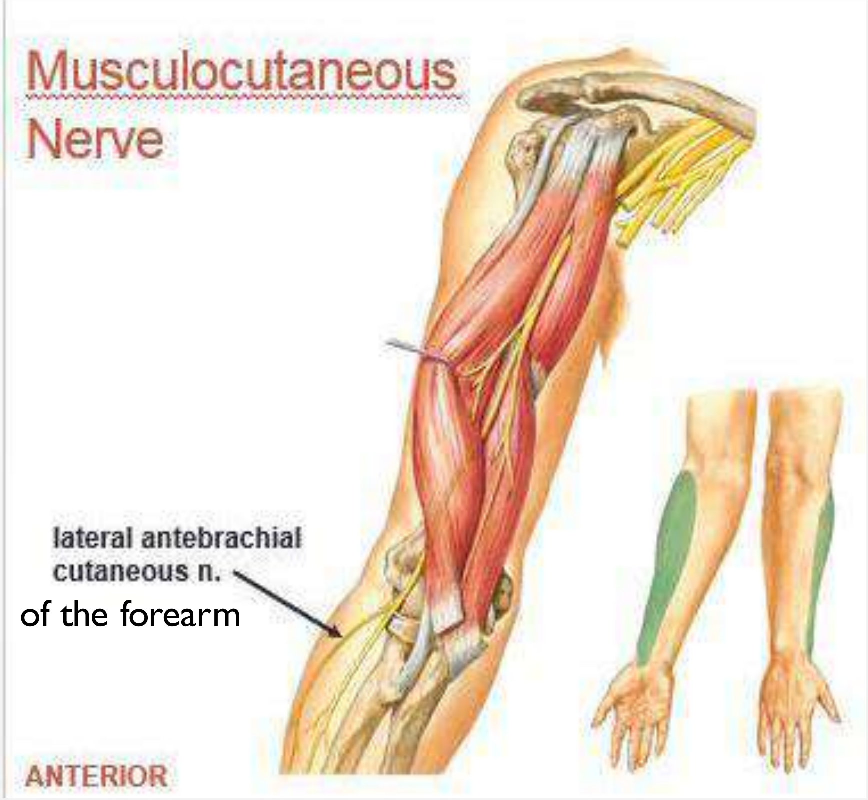
Ends: as the lateral cutaneous nerve of the forearm.

Branches:

■ **Muscular branches:**
to biceps, coracobrachialis, part of brachialis.

■ **Cutaneous branches:**
the lateral cutaneous nerve of the forearm:
supplies ‘skin of the front & lateral forearm down till the root of ‘thumb.

■ **Articular branches :** to the **elbow joint**



Median Nerve ~ In the arm

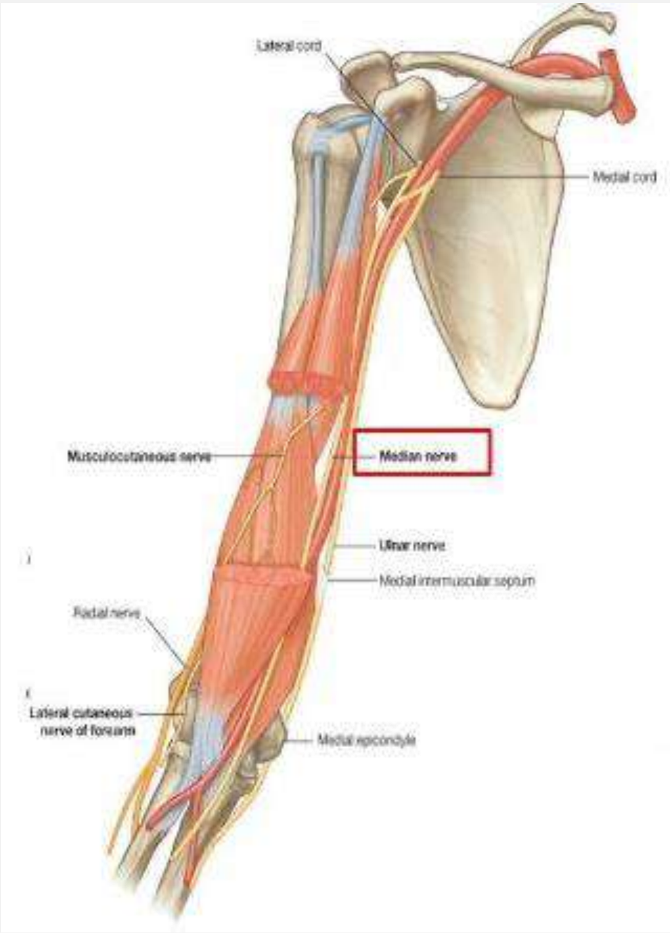
Begins:
in the axilla, arises from medial & lateral cords of 'brachial plexus.

Course:
It runs downward lateral to the brachial artery, then it crosses in front of middle part of 'brachial artery to be medial to it in the **cubital fossa**.

Branches:

NO branches in the arm

except vasomotor to brachial artery.
Articular to the elbow joint.



Ulnar Nerve ~ In the arm

Begins:

in the axilla, arises from medial cord of ‘brachial plexus.

Course and relations:

~ It descends medial to ‘brachial artery.

At the middle of ‘arm:

The ulnar nerve accompanied by the superior ulnar collateral artery,

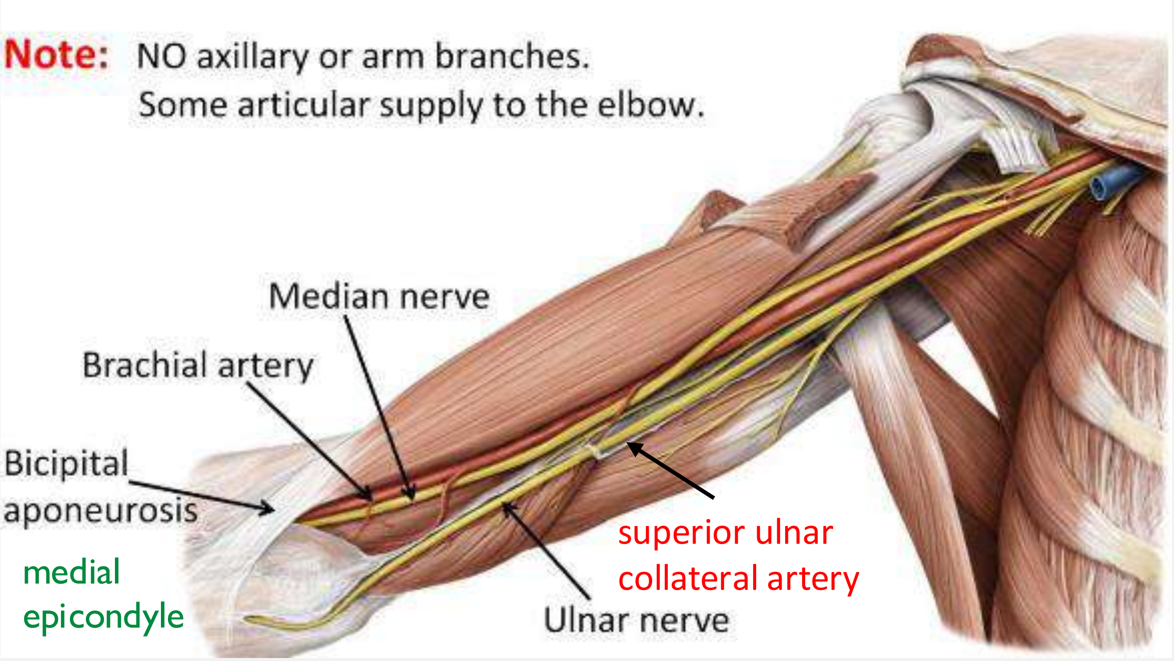
It enters the posterior compartment of the arm.

~ The nerve passes behind the medial epicondyle of the humerus.

Branches:

NO branches in the arm

Articular to the Elbow joint



Radial Nerve ~ In the arm

Begins:
in the axilla, arises from posterior cord of ‘brachial plexus.

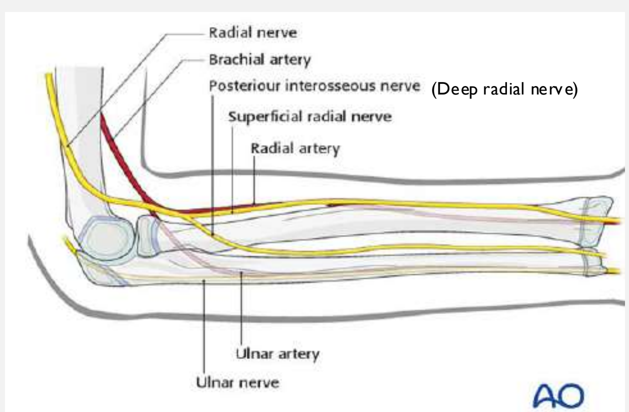
Course:
It leaves the axilla and immediately enters the posterior compartment.

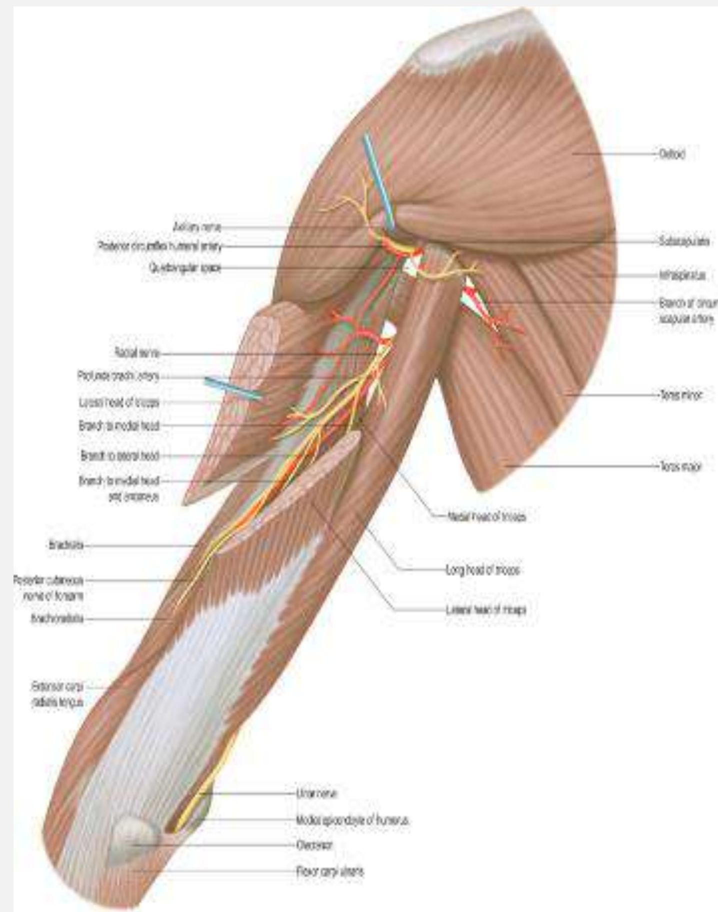
~ The radial nerve winds around in the spiral groove on the back of ‘humerus (accompanied by the profunda vessels).

~ In the lower part of the arm:
It enters the anterior compartment &
passes **in front** of ‘lateral epicondyle of ‘humerus.

Ends: At the level of the lateral epicondyle
by dividing into superficial and deep branches.

↓ Passes into ‘cubital fossa.





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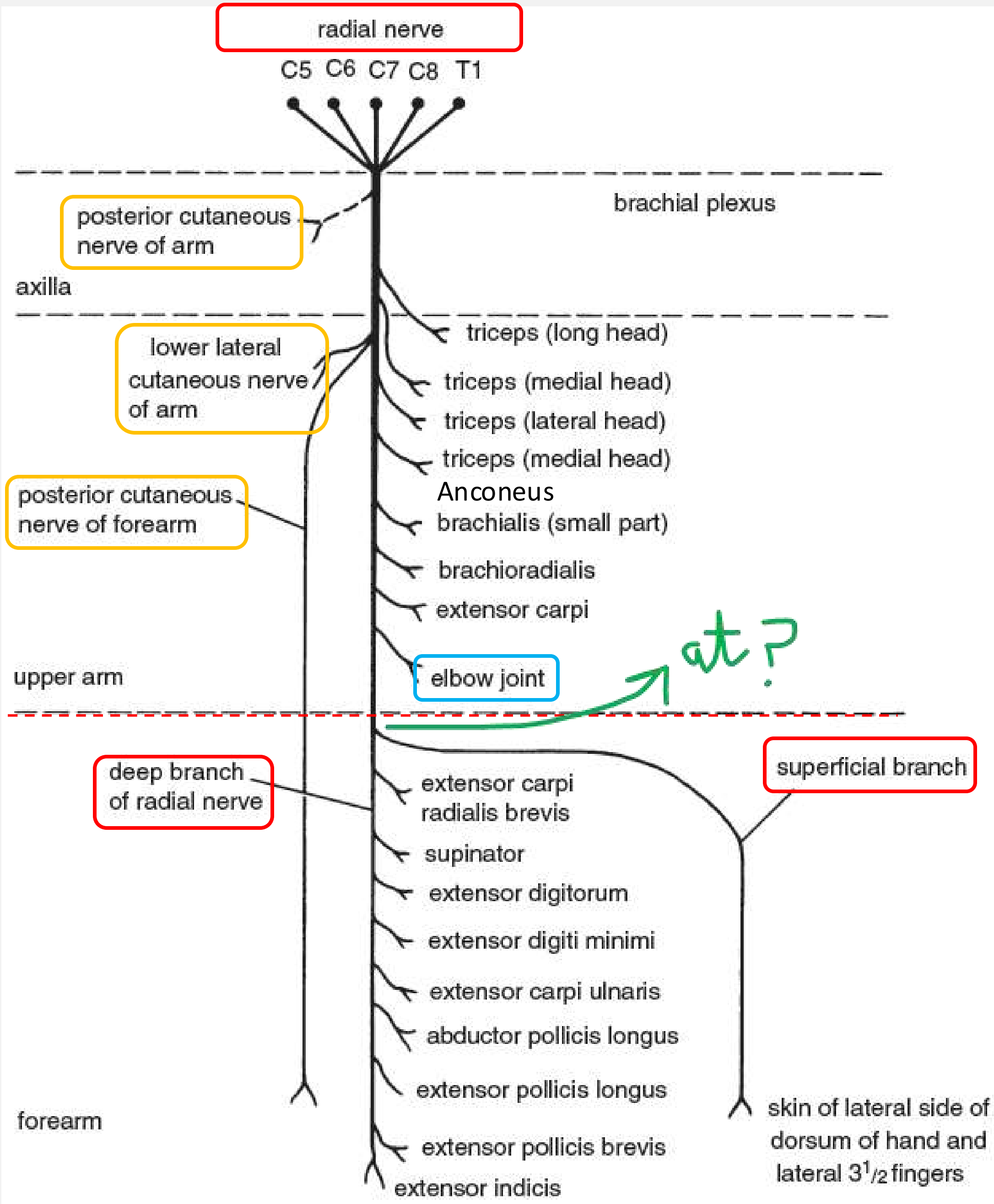
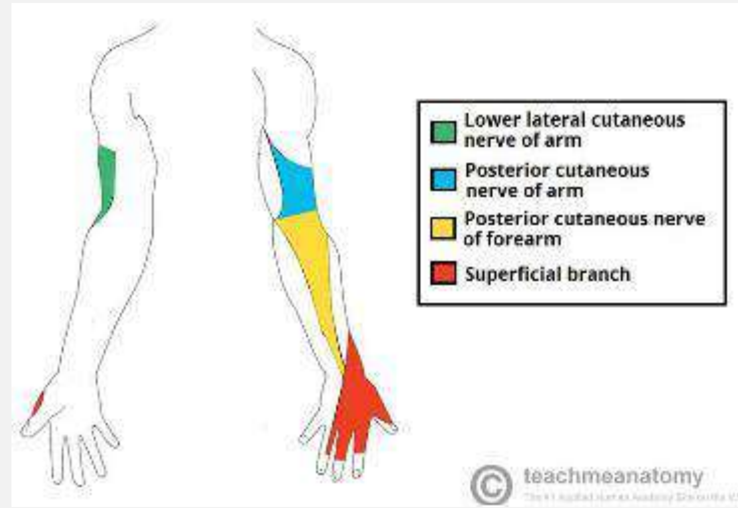
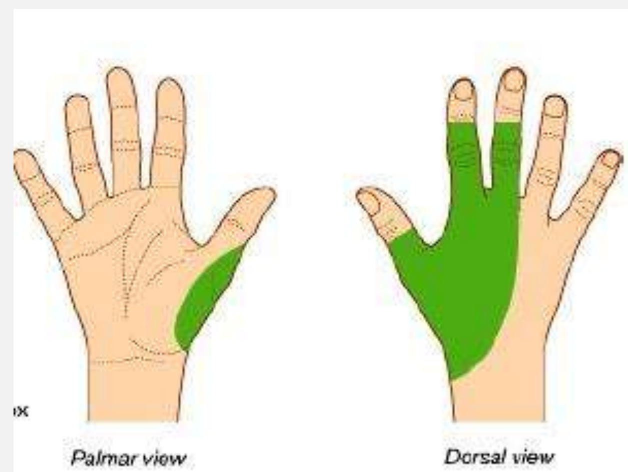


Remember :
the Upper lateral cutaneous nerve of the arm …. Branch of ?





Branches in axilla and arm:

■ In the **axilla** and **arm (spiral groove):**

~ **Muscular** branches to supply:

three heads of **Triceps,**

lateral part of **brachialis,** + **Anconeus, brachioradialis, ECRL.**

~ **Cutaneous** branches:

✓The lower lateral cutaneous nerve of the **arm**

supplies skin over ‘lateral & anterior aspects of 'lower part of ‘arm.

✓The posterior cutaneous nerve of the **arm.**

✓The posterior cutaneous nerve of the **forearm**

supplies back of forearm as far as the wrist.

■ **In the cubital fossa (at level of lateral epicondyle):**

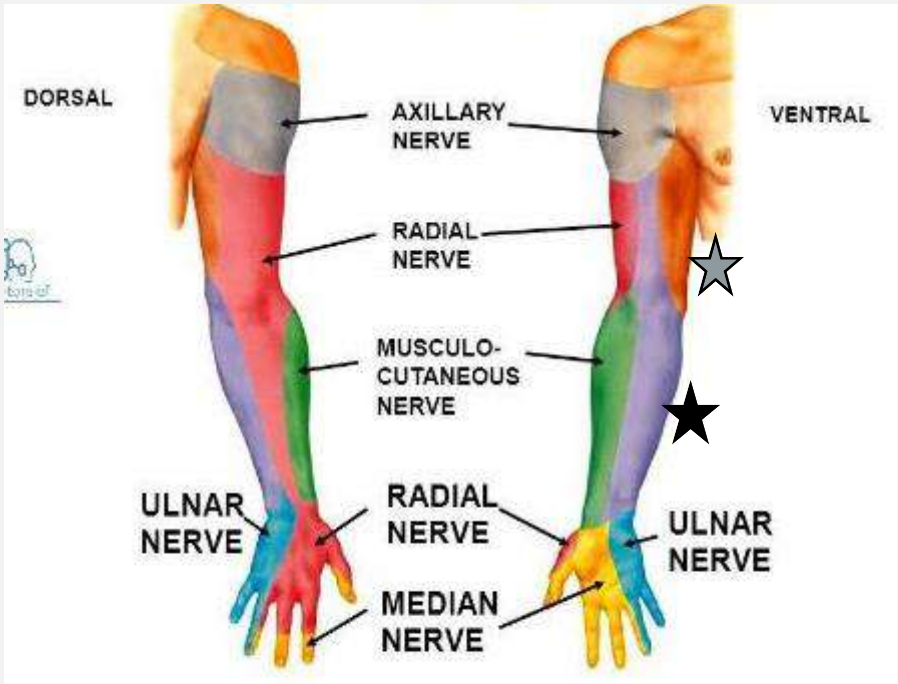
Deep branch/ **Posterior interosseous n.** (motor) to …..

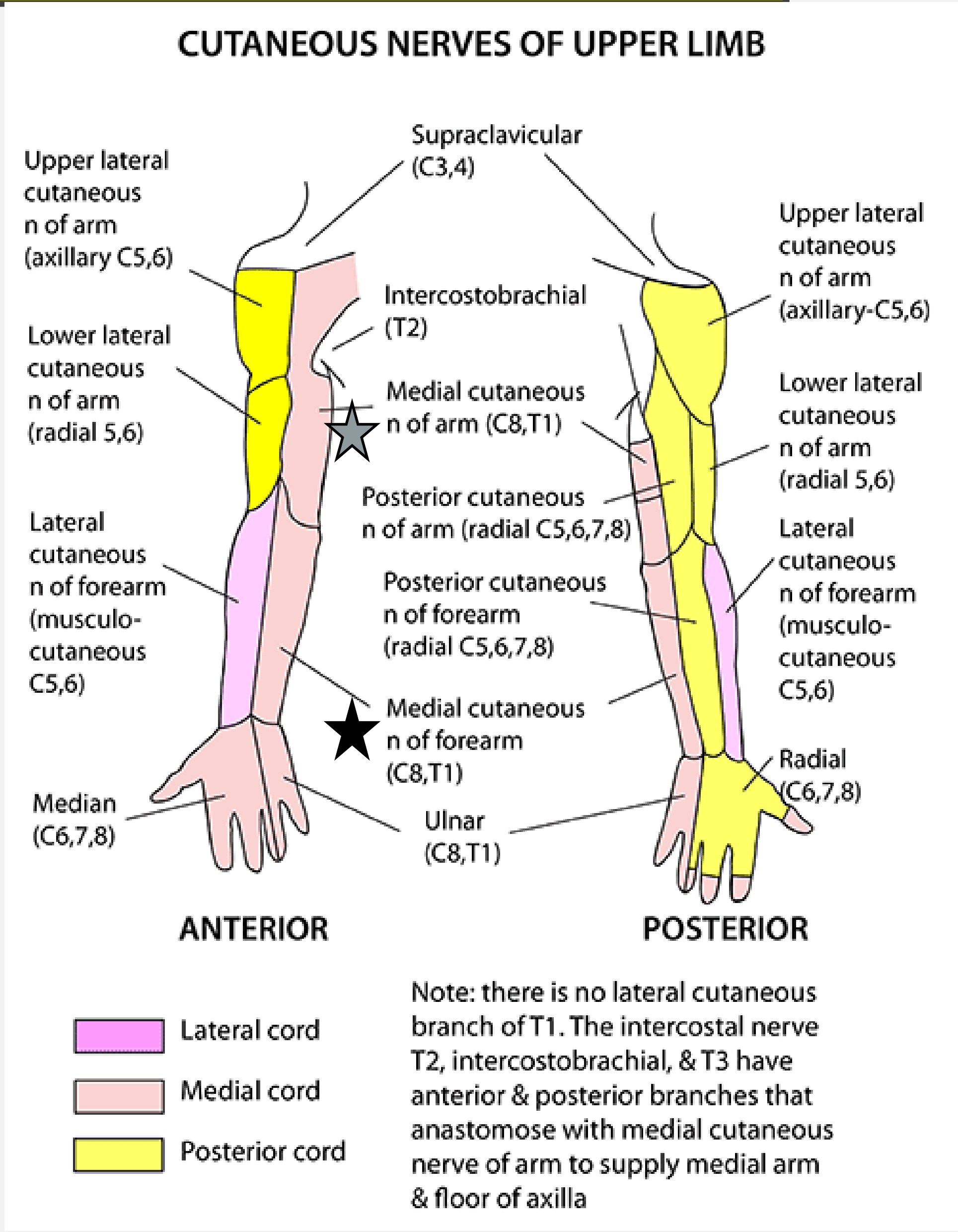
Superficial branch (sensory)to…..

■ **Articular branches:**

to the **elbow joint.**

SENSORY INNERVATION OF THE UPPER LIMB





SUPERFICIAL VEINS OF THE UPPER LIMB

They lie in the superficial fascia

- Cephalic v
- Basilic v
- Median cubital v

__It is used for :__

Intravenous injection..

Blood transfusion..

withdrawal of blood…

CEPHALIC VEIN

Begins:

Arises from lateral side of the dorsum – back of the hand.

It ascends on lateral side of 'forearm & arm.

Ends:

by piercing the clavipectoral fascia to join the axillary vein.

BASILIC VEIN

Begins:

Arises from medial side of the dorsum /back of the hand.

It ascends on medial side of 'forearm & arm.

Ends:

In the middle of the arm, by joining the

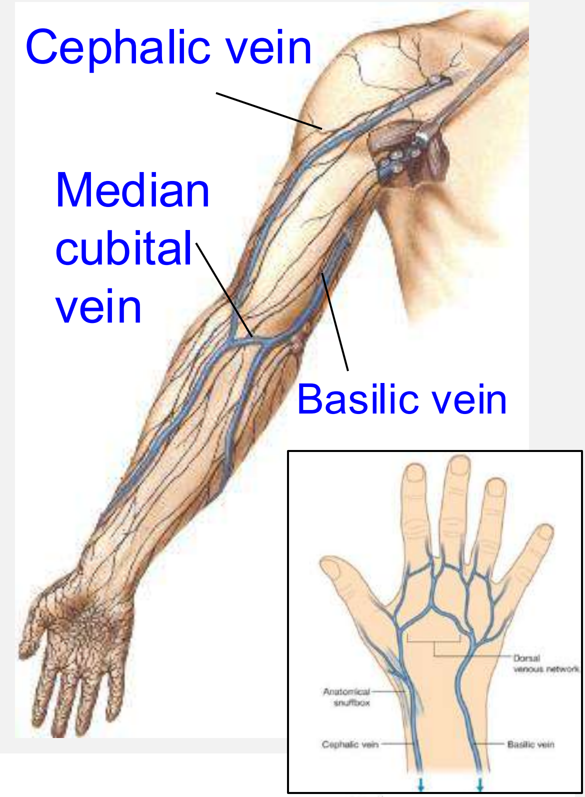
2 venae comitantes of the brachial artery

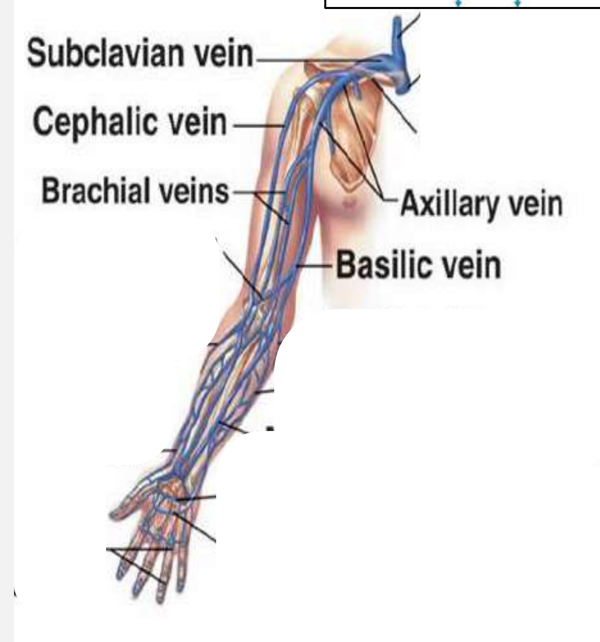
to form → the axillary vein.

MEDIAN CUBITAL VEIN

It communicates the <u>cephalic & basilic</u> veins on the roof of the cubital fossa.

It is separated from the brachial artery & median nerve by the bicipital aponeurosis.





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